Mathematical Derivations for the Cont–Xiong BSDEJ Framework

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Abstract

This note collects three mathematical contributions that underpin the computational programme of the dissertation. Section 1 derives the explicit jump-augmented backward stochastic differential equation (BSDEJ) representation of the Cont and Xiong [2024] multi-agent dealer market game — a translation not written down in the source paper but required for the application of deep BSDE methods. Section 2 invokes the quasi-linear PDE / FBSDE weak-solution correspondence of Feng, Wang, and Zhao [2018] to establish the well-posedness of the continuous-inventory diffusion limit of this BSDEJ. Section 3 formalises the three structural pathologies of neural-network approximations to the BSDEJ that are identified but not proven in the accompanying literature review: BatchNorm erasure, DeepSets representation collapse, and generator bypass via terminal-loss underdetermination.

1 BSDEJ representation of the Cont–Xiong game

Cont and Xiong [2024] formulate their dealer market as a stochastic differential game whose Nash equilibrium is characterised by a coupled system of Hamilton–Jacobi–Bellman (HJB) equations in their equation (28). They do not explicitly write this system as a backward stochastic differential equation with jumps (BSDEJ). This translation is a prerequisite for the application of the deep BSDE framework of Han, Jentzen, and E [2018] and its jump-augmented extension [Wang, Wang, Li, Wang, and Li, 2023], and is the starting point of this thesis's computational programme.

1.1 Setup

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \P)$ carrying two independent Poisson random measures $N^a(\mathrm{d}s, \mathrm{d}z)$ and $N^b(\mathrm{d}s, \mathrm{d}z)$ on $[0,T] \times \{1\}$ with $\mathcal{F}_t$-predictable intensities ν_t^a, ν_t^b. Let $q_t \in \{-Q, -Q+\Delta, \ldots, Q-\Delta, Q\}$ denote the dealer's inventory on the lattice $\mathcal{Q}$ of spacing Δ. The dealer controls bid and ask spreads $\delta_t^a, \delta_t^b \in [-1, \delta_{\max}]$. Inventory evolves as a pair of controlled Poisson jumps,

$$\mathrm{d}q_t = -\Delta\,\mathrm{d}N_t^a + \Delta\,\mathrm{d}N_t^b, \qquad q_0 = q \in \mathcal{Q}, \tag{1}$$

where N^a jumps when a sell order is executed against the dealer's ask and N^b jumps when a buy order is filled. The intensities depend on the dealer's own spread and on the average competitor spread m_t via the softmax kernel of Cont and Xiong [2024, eq. 58]:

$$\nu_t^a = \lambda_a\, f_a(\delta_t^a, m_t), \qquad \nu_t^b = \lambda_b\, f_b(\delta_t^b, m_t), \tag{2}$$

with

$$f_a(\delta, m) = \frac{1}{N}\,\sigma(-\delta)\,\frac{e^m}{1+e^{\delta+m}}, \tag{3}$$

where $\sigma(\cdot)$ is the logistic sigmoid and an analogous formula holds for f_b. The running cost is quadratic inventory penalty $\psi(q) = \phi q^2$; the terminal cost is $g(q) = -\phi q^2$. The dealer's objective is to maximise the expected discounted profit,

$$V(t,q) = \sup_{\delta^a,\delta^b} \mathbb{E}_{t,q}\left[\int_t^T e^{-r(s-t)}\Big(\delta_s^a \Delta\,\mathrm{d}N_s^a + \delta_s^b \Delta\,\mathrm{d}N_s^b - \psi(q_s)\,\mathrm{d}s\Big) + e^{-r(T-t)} g(q_T)\right]. \tag{4}$$

Standard dynamic-programming arguments [Carmona and Delarue, 2018] give the HJB equation, which under Cont and Xiong [2024]'s mean-field closure (i.e. m_t treated as given, exogenous quote aggregate) reads:

$$rV(t,q) - \partial_t V(t,q) = \max_{\delta^a,\delta^b} \mathcal{H}(t,q,\delta^a,\delta^b,V) - \psi(q), \qquad V(T,q) = g(q), \tag{5}$$

where the Hamiltonian is

$$\begin{aligned}\mathcal{H}(t,q,\delta^a,\delta^b,V) = \lambda_a\, f_a(\delta^a,m_t)[\delta^a\Delta + V(t,q-\Delta) - V(t,q)] \\ + \lambda_b\, f_b(\delta^b,m_t)\Big[\delta^b\Delta + V(t,q+\Delta) - V(t,q)\Big].\end{aligned} \tag{6}$$

(Boundary terms $V(t,q\pm\Delta) = V(t,q)$ apply at $q = \pm Q$ to reflect the inventory constraint.)

1.2 Derivation

Define the associated value, jump, and intensity processes along the controlled trajectory (q_t):

$$\begin{aligned} Y_t &:= V(t,q_t), \\ U_t^a &:= V(t,q_{t-}-\Delta) - V(t,q_{t-}), \\ U_t^b &:= V(t,q_{t-}+\Delta) - V(t,q_{t-}). \end{aligned} \tag{7}$$

U_t^a and U_t^b are the jumps experienced by Y at a sell-execution and buy-execution event respectively. Denote the compensated Poisson martingales by $\widetilde{N}_t^a = N_t^a - \int_0^t \nu_s^a \mathrm{d}s$ and similarly for $\widetilde{N}^b$.

Under $\delta_t^{a,\star}, \delta_t^{b,\star}$ achieving the maximum in (5), apply Itô's formula for semimartingales with jumps [Øksendal, 1998, Theorem III.7.32] to $Y_t = V(t,q_t)$:

$$\mathrm{d}Y_t = \partial_t V(t,q_t)\,\mathrm{d}t + U_t^a\,\mathrm{d}N_t^a + U_t^b\,\mathrm{d}N_t^b. \tag{8}$$

Decompose the jump terms into compensated martingale and drift:

$$U_t^a\,\mathrm{d}N_t^a = U_t^a\,\mathrm{d}\widetilde{N}_t^a + U_t^a\,\nu_t^a\,\mathrm{d}t, \tag{9}$$

and analogously for b. Substituting and collecting drift terms,

$$\mathrm{d}Y_t = \Big[\partial_t V(t,q_t) + U_t^a\nu_t^a + U_t^b\nu_t^b\Big]\,\mathrm{d}t + U_t^a\,\mathrm{d}\widetilde{N}_t^a + U_t^b\,\mathrm{d}\widetilde{N}_t^b. \tag{10}$$

The HJB (5) evaluated at $q = q_t$ and at optimal $\delta^\star$ gives

$$\partial_t V(t,q_t) = rV(t,q_t) + \psi(q_t) - \nu_t^{a,\star}\big[\delta_t^{a,\star}\Delta + U_t^a\big] - \nu_t^{b,\star}\Big[\delta_t^{b,\star}\Delta + U_t^b\Big],$$

where $\nu_t^{a,\star} := \lambda_a f_a(\delta_t^{a,\star}, m_t)$. Since the optimal intensities under the Nash-closure equal the process intensities, $\nu_t^{a,\star} = \nu_t^a$ and similarly for b, the $U^a\nu^a$ and $U^b\nu^b$ terms *cancel* between (10) and the HJB substitution. What remains is:

$$\mathrm{d}Y_t = \Big[rY_t + \psi(q_t) - \nu_t^{a,\star}\,\delta_t^{a,\star}\Delta - \nu_t^{b,\star}\,\delta_t^{b,\star}\Delta\Big]\,\mathrm{d}t + U_t^a\,\mathrm{d}\widetilde{N}_t^a + U_t^b\,\mathrm{d}\widetilde{N}_t^b. \tag{11}$$

The optimal quotes $\delta_t^{a,\star}, \delta_t^{b,\star}$ are determined by the first-order condition derived from (6); since f_a depends on δ^a only through the factor $\sigma(-\delta^a)/(1+e^{\delta^a+m_t})$ which is strictly log-concave in δ^a, the optimiser is unique and given implicitly by

$$\delta_t^{a,\star} = \arg\max_\delta\, f_a(\delta,m_t)(\delta\Delta + U_t^a), \qquad \delta_t^{b,\star} = \arg\max_\delta\, f_b(\delta,m_t)(\delta\Delta + U_t^b). \tag{12}$$

Writing (11) in backward form with terminal condition,

$$\boxed{-\mathrm{d}Y_t = F(t, q_t, Y_t, U_t^a, U_t^b)\,\mathrm{d}t - U_t^a\,\mathrm{d}\widetilde{N}_t^a - U_t^b\,\mathrm{d}\widetilde{N}_t^b, \qquad Y_T = g(q_T),} \tag{13}$$

with generator

$$F(t, q, y, u^a, u^b) = -ry - \phi q^2 + \lambda_a f_a(\delta^{a,\star}, m_t)\delta^{a,\star}\Delta + \lambda_b f_b(\delta^{b,\star}, m_t)\delta^{b,\star}\Delta, \tag{14}$$

where $\delta^{a,\star} = \delta^{a,\star}(u^a, m_t)$ and $\delta^{b,\star} = \delta^{b,\star}(u^b, m_t)$ are as in (12).

Proposition 1. *Under the Cont and Xiong [2024] intensity-control dealer-market model with mean-field quote closure $m_t = m(t, \mu_t)$, the value function $V(t, q)$ solving the HJB equation* (5) *admits a representation as the Y-component of the BSDEJ* (13) *with generator* (14) *and terminal condition $Y_T = -\phi q_T^2$. The diffusion coefficient Z vanishes identically because the forward SDE* (1) *has no Brownian component; the BSDEJ is therefore pure-jump.*

Proof. Immediate from the derivation above. Uniqueness of (Y, U^a, U^b) in the space $\mathbb{H}^2 \times \mathbb{H}^2(\widetilde{N}^a) \times \mathbb{H}^2(\widetilde{N}^b)$ follows from the BSDEJ well-posedness result of Barles, Buckdahn, and Pardoux [1997] and Tang and Li [1994], given the Lipschitz regularity of the generator F in (y, u^a, u^b) (which follows from smoothness of f_a, f_b and strict log-concavity of the FOC). □

Remark 2. Proposition 1 highlights a structural feature of the CX model: its BSDEJ is degenerate in the sense that the diffusion coefficient Z_t vanishes identically. The canonical deep BSDE method of Han et al. [2018] parameterises Z by a neural network; in our setting, the network instead parameterises (U^a, U^b), the jump coefficients. Recovering a non-degenerate deep BSDE with $Z \neq 0$ requires replacing the discrete Poisson inventory dynamics with a diffusion approximation, to which we turn in Section 2.

2 Existence in the continuous-inventory limit

The BSDEJ (13) is the exact representation of the discrete-inventory CX game. For the deep BSDE method to learn the gradient process $Z_t = \sigma_q\,\partial_q V$ of its canonical formulation, we pass to a continuous-inventory diffusion approximation of (1).

2.1 Diffusion approximation

In the regime of high execution intensities and short inter-arrival times — the Poisson central-limit regime — the controlled jump process (1) is approximated in law by a controlled diffusion:

$$\mathrm{d}q_t = \mu_q(q_t, \delta_t^a, \delta_t^b)\,\mathrm{d}t + \sigma_q(q_t, \delta_t^a, \delta_t^b)\,\mathrm{d}W_t, \tag{15}$$

with drift and diffusion coefficients

$$\begin{aligned} \mu_q &= \Delta\Big[\lambda_b f_b(\delta^b, m) - \lambda_a f_a(\delta^a, m)\Big], \\ \sigma_q^2 &= \Delta^2\Big[\lambda_a f_a(\delta^a, m) + \lambda_b f_b(\delta^b, m)\Big]. \end{aligned} \tag{16}$$

Under (15), the associated backward equation acquires a Brownian martingale term:

$$-\mathrm{d}Y_t = \widetilde{F}(t, q_t, Y_t, Z_t)\,\mathrm{d}t - Z_t\,\mathrm{d}W_t, \qquad Y_T = g(q_T), \tag{17}$$

where

$$\widetilde{F}(t, q, y, z) = -ry - \phi q^2 + \lambda_a f_a(\delta^{a,\star}, m)\delta^{a,\star}\Delta + \lambda_b f_b(\delta^{b,\star}, m)\delta^{b,\star}\Delta, \tag{18}$$

and the optimal quotes now satisfy the modified FOC

$$\delta^{a,\star} = \arg\max_{\delta} f_a(\delta, m)(\delta\Delta - (Z/\sigma_q)\Delta), \qquad \delta^{b,\star} = \arg\max_{\delta} f_b(\delta, m)(\delta\Delta + (Z/\sigma_q)\Delta), \tag{19}$$

using the identity $\partial_q V = Z/\sigma_q$ from the martingale representation of the Brownian-only FBSDE.

2.2 Invocation of Feng–Wang–Zhao

Feng et al. [2018] prove existence of weak solutions to quasi-linear parabolic PDEs of the form

$$\partial_t u + \tfrac{1}{2}\mathrm{tr}(\sigma\sigma^\top \nabla^2 u) + b^\top \nabla u + f(t, x, u, \sigma^\top \nabla u) = 0 \tag{20}$$

via the associated FBSDE representation, under the following hypotheses on the coefficients (b, σ, f): local Lipschitz continuity in $(x, u, \nabla u)$, linear growth, and suitable compactness of the terminal condition. Their main result, [Feng et al., 2018, Theorem 3.3], establishes that if these hypotheses hold, then the PDE (20) admits a weak solution in the space $L^2(0, T; H^1) \cap C([0, T]; L^2)$, and the solution coincides with the decoupling field $u(t, x) = Y_t^{t,x}$ of the FBSDE.

Assumption 3. The market parameters satisfy:

1. $\lambda_a, \lambda_b > 0$ bounded;
2. $r, \phi > 0$; $\Delta \in (0, \infty)$;
3. $m \in [-M, M]$ for some $M < \infty$;
4. $\delta \in [-1, \delta_{\max}]$ for some $\delta_{\max} < \infty$.

Lemma 4. *Under Assumption 3, the coefficients $(b, \sigma, f) = (\mu_q, \sigma_q, \widetilde{F})$ of the diffusion system (15)–(18) are locally Lipschitz in (q, y, z) and satisfy linear-growth bounds.*

Proof. The execution kernel $f_a(\delta, m)$ defined in (3) is C^∞ in (δ, m), and its partial derivatives are bounded on the admissible range of Assumption 3(3)–(4). Hence μ_q and σ_q are bounded and smooth in (δ^a, δ^b, m). The optimal quotes $\delta^{a,\star}(z, m)$ are determined by the unique zero of the strictly concave objective in (19); implicit-function arguments give local Lipschitz continuity in (z, m). Substituting back, $\widetilde{F}(t, q, y, z)$ inherits local Lipschitz continuity in (q, y, z), linear growth, and boundedness of the y-derivative by r. □

Corollary 5. *Under Assumption 3, the continuous-inventory BSDE (17)–(18) admits a unique solution $(Y, Z) \in \mathbb{H}^2 \times \mathbb{H}^2$ for each initial condition $q_0 \in [-Q, Q]$, and the associated value function $v(t, q) := Y_t^{t,q}$ is a weak solution of the quasi-linear PDE*

$$\partial_t v + \mu_q(q, \delta^\star)\partial_q v + \tfrac{1}{2}\sigma_q^2(q, \delta^\star)\partial_q^2 v + \widetilde{F}(t, q, v, \sigma_q \partial_q v) = 0, \qquad v(T, q) = g(q). \tag{21}$$

Proof. Combine Lemma 4 with [Feng et al., 2018, Theorem 3.3]. □

Remark 6. For the discrete-inventory BSDEJ (13), existence of (Y, U^a, U^b) does not require Feng–Wang–Zhao; it follows from the pure-jump BSDEJ theory of Barles et al. [1997] and Tang and Li [1994] directly. The point of Corollary 5 is to cover the diffusion extension, which is what the genuine deep BSDE method requires.

3 Structural pathologies of deep BSDE approximations

The literature review identifies three failure modes of neural-network parameterisations of the BSDEJ (13) whose consequence is approximation to the wrong HJB system. We formalise each.

3.1 Loss of μ-measurability via BatchNorm

Let B denote the minibatch size during stochastic gradient descent, and consider a feature $x \in \mathbb{R}^B$ supplied to a BatchNorm layer at training time. Let $\mu_B := \frac{1}{B}\sum_i x_i$ and $\sigma_B^2 := \frac{1}{B}\sum_i (x_i - \mu_B)^2$, and write $\mathrm{BN}(x) := (x - \mu_B \mathbf{1})/(\sigma_B + \varepsilon)$ for a numerical-stability constant $\varepsilon > 0$.

Lemma 7. *If $x = c\,\mathbf{1}$ for some constant $c \in \mathbb{R}$, then $\mathrm{BN}(x) = \mathbf{0}$ identically.*

Proof. Direct computation: $\mu_B = c$, $\sigma_B^2 = 0$, and $\mathrm{BN}(x) = (c - c)/\varepsilon \cdot \mathbf{1} = \mathbf{0}$. □

Proposition 8 (Loss of μ-measurability)**.** *Let $\Phi_\theta : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^k$ denote a neural network parameterising the value function $V(t, x, \mu_t) \approx \Phi_\theta(t, x, m_t)$ of a mean-field game, where $m_t \in \mathbb{R}$ is a broadcast population statistic. If Φ_θ applies a BatchNorm layer to the input coordinate carrying m_t, then the output of Φ_θ is $\sigma(x, t)$-measurable independently of m_t; equivalently, the approximation satisfies $\partial_{m_t}\Phi_\theta \equiv 0$. The discretised mean-field HJB therefore collapses to the single-agent HJB.*

Proof. m_t is broadcast constant across the minibatch dimension by construction (it is a single scalar for the entire population). Lemma 7 gives that the BatchNorm output of the m_t coordinate is the zero vector for every minibatch. Subsequent layers receive no signal from m_t; by the chain rule, $\partial_{m_t}\Phi_\theta = 0$ on the training distribution. The resulting $\Phi_\theta(t, x)$ solves the Bellman equation without the mean-field coupling, i.e. the single-agent HJB. □

3.2 Representation collapse of symmetric DeepSets

A DeepSets parameterisation [Zaheer, Kottur, Ravanbakhsh, Póczos, Salakhutdinov, and Smola, 2017] of a permutation-invariant function $\mu \mapsto \rho(\mu)$ takes the form $\Phi_\theta(\{x_1, \ldots, x_K\}) = \rho_\theta\Big(\sum_{i=1}^K \phi_\theta(x_i)\Big)$.

Proposition 9 (Degenerate collapse)**.** *Suppose ϕ_θ and ρ_θ are initialised with parameters drawn i.i.d. from a distribution symmetric under coordinate permutations, and suppose the training distribution consists of K-tuples with $x_i \equiv x$ for all i (i.e. symmetric inputs). Then the gradient of any symmetric loss $\mathcal{L}(\Phi_\theta(\cdot))$ with respect to θ preserves the symmetry of the initialisation throughout gradient descent; in particular, training converges to a stationary point at which $\partial_\mu \rho_\theta$ is zero on the empirical measure of the training data.*

Proof. The loss landscape is invariant under the group of permutations acting on the input coordinates. The initialisation is invariant under this group action by assumption. Gradient flow of an invariant loss from an invariant initialisation preserves the invariance at every iterate. In particular the learned map has no gradient in the direction that distinguishes the uniform empirical measure $\mu = \delta_x$ from perturbations of it that would break the permutation symmetry. □

Remark 10. Proposition 9 is, like Proposition 8, a statement about loss of measurable dependence on μ — but the mechanism is a degeneracy of the minimiser rather than an active erasure. In practice the two are operationally indistinguishable.

3.3 Terminal-loss underdetermination (generator bypass)

The canonical deep BSDE training objective for (13) is the empirical terminal loss

$$\widehat{\mathcal{L}}_B(\theta) = \frac{1}{B}\sum_{i=1}^{B}\left|Y_T^{(i)}(\theta) - g(q_T^{(i)})\right|^2, \tag{22}$$

where the $(Y_T^{(i)})$ are generated by forward propagation of the discretised BSDEJ under parameter θ on a sample of B paths $(q_T^{(i)})$. The Han et al. [2018] method treats Y_0 and the jump-coefficient network (U_θ^a, U_θ^b) as jointly learnable.

Proposition 11 (Underdetermination of terminal loss)**.** *Let $\mathcal{U}^\star = \{(u^{a,\star}, u^{b,\star}) \in \mathbb{H}^2(\widetilde{N}^a) \times \mathbb{H}^2(\widetilde{N}^b)\}$ denote the unique jump coefficients of the BSDEJ (13) in the continuum limit. Fix an empirical batch $(q_T^{(i)})_{i=1}^B$ and consider the forward BSDEJ propagation restricted to this batch. The set of parameters θ for which $\widehat{\mathcal{L}}_B(\theta) < \epsilon$ (for any $\epsilon > 0$) contains an open neighbourhood of*

$\theta^\star := \theta(\mathcal{U}^\star)$ but is not reduced to it. In particular, there exist $\theta \neq \theta^\star$ with $\widehat{\mathcal{L}}_B(\theta)$ arbitrarily small but extracted optimal quotes $\delta^{a,\star}(U^a_\theta), \delta^{b,\star}(U^b_\theta)$ differing substantially from the Bellman-optimal $\delta^{a,\star}(u^{a,\star}), \delta^{b,\star}(u^{b,\star})$.

Proof sketch. Parameter count. The network Φ_θ has $\dim(\theta)$ parameters (typically 10^4 or more for a width-128 MLP); the empirical terminal loss is a single scalar constraint averaged over B paths (typically $B \leq 10^3$). The preimage $\widehat{\mathcal{L}}_B^{-1}([0,\epsilon))$ in parameter space is therefore generically a submanifold of codimension at most B within the $\dim(\theta)$-dimensional parameter space, whose intrinsic dimension is at least $\dim(\theta) - B$. In the limit $B \to \infty$ with fixed θ, the law of large numbers pins this preimage to $\{\theta^\star\}$ (via the uniqueness part of the martingale representation theorem for (13)); but for finite B, the preimage has positive Lebesgue measure in parameter space.

Explicit counterexample. Consider a one-step discretisation of (13) on a two-point state space $q \in \{-\Delta, 0\}$ with a single Poisson jump time τ. Write $Y_1 = Y_0 - F\Delta t + U^a(N^a - \nu^a \Delta t)$. On any sample path, the equality $Y_1 = g(q_1)$ is a linear equation in the two unknowns (Y_0, U^a) (treating ν^a, F as given). A single sample path gives one equation in two unknowns; infinitely many (Y_0, U^a) satisfy it. Only aggregation over infinitely many paths — exploiting the strict concavity of the quadratic loss — pins (Y_0, U^a) to the Bellman values. This underdetermination propagates to the discretised BSDEJ on $\mathcal{Q}$ for any finite B. □

Corollary 12 (Non-Bellman fixed points of MV-fictitious play)**.** *When the discretised BSDEJ is embedded in the outer McKean–Vlasov fictitious-play iteration* $\text{avg_comp}^{(k+1)} = (1-\eta)\text{avg_comp}^{(k)} + \eta\, \overline{\delta^\star(U^{(k)}_\theta)}$*, Proposition 11 implies that the iteration can converge to fixed points* $\text{avg_comp}^{(\infty)}$ *at which the extracted quote average* $\overline{\delta^\star(U_\theta)}$ *equals* $\text{avg_comp}^{(\infty)}$ *(so the outer FP is consistent) but neither U_θ nor $\overline{\delta^\star(U_\theta)}$ corresponds to the Nash equilibrium of the CX game.*

Proof. Immediate from Proposition 11 combined with the contractivity of damped fictitious-play. The outer iteration is indifferent to which element of $\widehat{\mathcal{L}}_B^{-1}([0,\epsilon))$ is selected by the inner optimiser; any such element produces a self-consistent fixed point of the outer map, whether or not it coincides with the Nash equilibrium. □

Corollary 12 is the precise statement of the "spurious fixed point" phenomenon observed empirically in this thesis's MV-fictitious-play experiments. The overnight queue of regularisation schemes (v4–v8 in the experimental programme) targets Proposition 11 directly: each introduces a prior on θ that shrinks the effective dimension of the preimage, pushing the optimiser toward $\theta^\star$.

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